Incorporating DRAG into Space-Curve Quantum Control and Leakage-Aware BARQ

Technical design note for the `qurveros` package

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Abstract

This note derives how Derivative Removal by Adiabatic Gate (DRAG) can be combined with space-curve quantum control (SCQC) and the Bézier ansatz for robust quantum control (BARQ), as implemented in `qurveros`. The central result is that a DRAG correction applied after a completed BARQ design is inexpensive, but it does not automatically preserve the exact geometric robustness conditions. It preserves the computational gate only perturbatively in $1/(\alpha T)$ and generally introduces mixed errors of order $(\delta_z T)^2/(\alpha T)^2$. A leakage-aware BARQ objective can instead impose four real first-order leakage constraints while recomputing dephasing and amplitude sensitivities under the corrected qutrit evolution. The note also records the associated boundary conditions, objective functions, spectral and smoothness extensions, the relation to a perturbative $\mathbb{C}^2$ leakage curve, and the scaling that can reasonably be expected before a numerical campaign.

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1 Scope

This note keeps the existing single-qubit SCQC and BARQ constructions and asks what changes when the same pulses drive a weakly anharmonic three-level system. DRAG is a pulse correction for that setting [1]. It is applied after the curve has been converted to fields. Sections 2 and 2.2 summarize the geometric formalism first. Section 3 fixes every symbol used later and records where it differs from `qurveros`, from Motzoi et al., and from the SCQC papers. Only then does the note introduce leakage and DRAG.

2 Single-qubit SCQC, with and without BARQ

2.1 Single-qubit SCQC without BARQ

The single-qubit drive used by Barnes et al. [7], including quasistatic dephasing, is

$$\begin{aligned}\frac{H(t)}{\hbar} = \frac{\Omega(t)}{2}\big(\cos\Phi(t)\,\sigma_x + \sin\Phi(t)\,\sigma_y\big) \\ + \frac{\Delta_c(t)+\delta_z}{2}\sigma_z.\end{aligned} \tag{1}$$

Here $\sigma_x, \sigma_y, \sigma_z$ are the Pauli operators on $\{|0\rangle, |1\rangle\}$, $\Omega(t)$ is the Rabi frequency, $\Phi(t)$ is the drive phase, and $\Delta_c(t)$ is the control detuning from the qubit transition. The coefficient δ_z is the quasistatic noise. It is constant during one gate but unknown, and it has the same units as Δ_c. Barnes et al. write the noise operator as $\delta H = \sigma_z$ and carry its small coefficient implicitly [7]. The factor $\frac{1}{2}$ in Eq. (1) matches the `qurveros` convention $H_n = \delta_z\sigma_z/2$. Because this factor multiplies the whole error integral, it does not change the closure or vanishing-area conditions. In the package, δ_z is the additive shift used by `static_dephasing_experiment`.

Let $U_c(t)$ be the evolution generated by the first three terms only, with $\mathrm{i}\hbar\dot{U}_c = H_c U_c$ and $U_c(0) = I$. The first-order error is the curve

$$\int_0^t U_c^\dagger(t')\sigma_z U_c(t')\,\mathrm{d}t' = \mathbf{r}(t)\cdot\boldsymbol{\sigma} = x(t)\sigma_x + y(t)\sigma_y + z(t)\sigma_z. \tag{2}$$

This is Eq. (7) of the review [7]. The vector $\mathbf{r}(t) = (x, y, z)$ is an ordinary curve in three-dimensional Euclidean space. Its curvature κ and torsion τ_c determine the controls:

$$\kappa(t) = \Omega(t), \qquad \tau_c(t) = \dot{\Phi}(t) - \Delta_c(t). \tag{3}$$

Thus a curve fixes the envelope and only the combination $\dot{\Phi} - \Delta_c$. A frame rotation about z can move the torsion into either the phase or the detuning. That is why one curve has several control modes. In `qurveros` they are selected by `control_mode`:

- `XY`: resonant drive, $\Delta_c = 0$, and Φ is the integral of the torsion;

- `XZ`: $\Phi = 0$ and $\Delta_c = -\tau_c$;
- `TTC`: a constant detuning compensates the total torsion, and the phase absorbs $\Delta_c t$;
- `resTTC`: the same phase as `TTC`, but the detuning is then set back to zero.

The package also multiplies Ω by a sign that flips at singular inflection points, so the stored quadratures stay continuous when the Frenet binormal reverses. The curvature in `frenet_dict` is the geometric κ, before that sign and before multiplication by the gate duration.

The gate implemented by the curve is read from the moving frame at the two endpoints, up to the final phase gauge. `qurveros` stores that gate as the 3×3 adjoint matrix `adj_curve`. No optimization is required for this step: any sufficiently smooth curve, for example the circle and Bessel examples, can be passed to `SpaceCurve`.

The robustness conditions used by the package are geometric constraints on $\mathbf{r}$. With T the duration, equivalently the length of a unit-speed curve:

- *Closure*, $\mathbf{r}(T) = \mathbf{r}(0)$. The first-order dephasing generator vanishes. The code reports this, normalized by T, as `closed_test`.
- *Vanishing projected area* of $\mathbf{r}$. This cancels the second-order dephasing term [7]. The code reports `curve_area_test`.
- *Vanishing area of the tantrix*, the curve traced by the unit tangent. This is the first-order condition for a multiplicative error in Ω. The code reports `tantrix_area_test`.
- *Curve flatness index*, the normalized energy $\int |\mathbf{r}|^2 \, \mathrm{d}s/T^3$. It estimates infidelity under $1/f^2$ noise and is reported as `CFI`.

Without BARQ these conditions are targets for whoever designs the curve. They are not built into an arbitrary `SpaceCurve`.

2.2 Single-qubit SCQC with BARQ

BARQ restricts the same three-dimensional curve to a Bézier parametrization. In `qurveros` the corresponding object is `BarqCurve`. The position is a Bézier curve on the fixed interval $s \in [0, 1]$,

$$\begin{aligned} \mathbf{r}(s) &= \sum_{i=0}^{n} B_{i,n}(s)\, \mathbf{W}_i, \\ B_{i,n}(s) &= \binom{n}{i} s^i (1-s)^{n-i}. \end{aligned} \tag{4}$$

with control points $\mathbf{W}_i \in \mathbb{R}^3$. The Bernstein weights $B_{i,n}$ are not a drive amplitude. Two structural choices are imposed before any numerical search.

Closure and a vanishing endpoint envelope. The first and last control points are set to the origin. The next points are arranged so that the curvature, and therefore Ω, vanishes at both endpoints. In the code this arrangement is `point_gate_fixing`.

Gate fixing. The same endpoint points are aligned with the adjoint representation of the target unitary. Consequently every later movement of the free interior points preserves the ideal qubit gate and the closure condition, to the accuracy of the two-level theory. The free parameters are the interior points, a dictionary of positive scale factors called the point-gate-fixing parameters, and one angle θ_B. The angle θ_B is the binormal angle at the final endpoint. It is an input to the constant-detuning calculation in `TTC` mode; it is not an anharmonicity and it is not the DRAG coefficient introduced below.

An optional point-relation function can constrain the interior points further. If it is omitted, those points pass through unchanged. The numerical search then minimizes a weighted sum of losses such as tantrix area, curve area, gate duration, peak curvature, curve flatness, and the

`TTC` detuning. This is ordinary single-qubit SCQC with part of the geometry fixed analytically. `qurveros` does not implement a multi-qubit version of this parametrization.

2.3 Multi-qubit and multilevel SCQC

The same error-curve idea extends beyond one qubit [6, 7], but the curve is no longer three-dimensional and BARQ does not automatically extend with it. For a control Hamiltonian H_c and a noise Hamiltonian δH, the review defines the normalized noise direction by $Q_{\mathrm{B}} = \delta H/|\delta H|$ and assumes $\{H_c, Q_{\mathrm{B}}\} = 0$. The subscript distinguishes this operator from the leakage projector Q introduced in Section 3. Barnes et al. call the noise direction Q. The leading-order error is again an integrated interaction-picture operator,

$$\int_0^t U_c^\dagger(t') Q_{\mathrm{B}} U_c(t')\,\mathrm{d}t' = \sum_{\alpha=1}^{d_E} x_\alpha(t)\, V_\alpha, \tag{5}$$

which is their Eq. (21). The operators V_α are the distinct directions generated by conjugating Q_{B}, and d_E is the dimension of the resulting Euclidean curve. It is determined by the Lie algebra generated by H_c and Q_{B}, not chosen freely. A single qubit with dephasing returns $d_E = 3$. The two-qubit Ising example worked out by Buterakos, Das Sarma, and Barnes has $d_E = 6$, and in that special case the curve splits into two coupled three-dimensional curves [6].

The controls are no longer a single curvature and torsion. An orthonormal Frenet–Serret frame $\mathbf{e}_n(t)$ in d_E dimensions has generalized curvatures κ_n obeying

$$\dot{\mathbf{e}}_n = -\kappa_{n-1}\mathbf{e}_{n-1} + \kappa_n \mathbf{e}_{n+1}, \tag{6}$$

with κ_1 the ordinary curvature and κ_2 the ordinary torsion when $d_E = 3$. The review gives a recursion that identifies these curvatures with nested commutators and time derivatives of H_c. Reading a pulse from a curve is therefore still possible. Designing the curve is harder: a physical Hamiltonian often requires some κ_n to be constant or to be functions of one another. The review states that a general method for imposing those constraints in dimension greater than three is open, and that the single-qubit vanishing-area condition has not been extended to the higher-dimensional curve. Closure, $\mathbf{r}(T) = \mathbf{r}(0)$, remains the condition for cancelling the leading Magnus term.

Two consequences matter for the rest of this note. First, multi-qubit SCQC without BARQ already exists as this curvature construction, but the gate must still be fixed by matching the endpoint frame, and that step is not automated by `qurveros`. Second, there is no BARQ-style Bézier gate-fixing theorem in the package, or in the references above, for a generic multi-qubit or qutrit curve. Leakage in the review is motivation for leaving the Bloch sphere. It is not given a separate DRAG calculation. The leakage conditions derived below are perturbative additions to the single-qubit theory, not a replacement for Eq. (5).

3 Notation and differences from the references

Time t runs from 0 to the gate duration T. The normalized time used throughout `qurveros` is

$$s = \frac{t}{T} \in [0, 1]. \tag{7}$$

A dot is $\mathrm{d}/\mathrm{d}t$ and a prime is $\mathrm{d}/\mathrm{d}s$. The computational projector and the leakage projector on a qutrit are

$$P = |0\rangle\langle 0| + |1\rangle\langle 1|, \qquad Q = I - P = |2\rangle\langle 2| \tag{8}$$

in the three-level model. For a larger truncation, Q is the projector onto every state outside $\{|0\rangle, |1\rangle\}$. This Q is not the noise direction Q_{B} of Section 2.3.

The driven three-level ladder, in the frame rotating at the carrier and after the rotating-wave approximation, is

$$\begin{aligned}\frac{H_3(t)}{\hbar} &= \delta(t)|1\rangle\langle 1| + \big(2\delta(t)+\alpha\big)|2\rangle\langle 2| \\ &\quad + \frac{1}{2}\Big\{\varepsilon(t)\big(|0\rangle\langle 1| + \lambda|1\rangle\langle 2|\big) + \text{h.c.}\Big\}.\end{aligned} \tag{9}$$

Each new symbol has a specific meaning:

- $|0\rangle, |1\rangle, |2\rangle$ are the lowest three eigenstates of the undriven anharmonic oscillator.
- ω_{01} and ω_{12} are their transition angular frequencies, in radians per unit time. They are properties of the device, not pulse fields.
- $\alpha = \omega_{12} - \omega_{01}$ is the anharmonicity. For a transmon, $\alpha < 0$. Motzoi et al. call this quantity Δ, with $\Delta = \omega_2 - 2\omega_1$ in their lab-frame numbering [1]. Barnes et al. use $\Delta(t)$ for the control detuning. This note never uses Δ for the anharmonicity; the control detuning is Δ_c in Eq. (1) and δ in Eq. (9).
- $\delta(t)$ is the instantaneous detuning of the carrier below the $|0\rangle \leftrightarrow |1\rangle$ transition, expressed as an angular frequency. Motzoi et al. call the constant part of this detuning δ_1. On the qubit subspace and at $\lambda = 0$, $\delta(t)$ is exactly $\Delta_c(t)$.
- $\varepsilon(t) = \Omega_x(t) + \mathrm{i}\Omega_y(t)$ is the complex envelope. Matching Eq. (1) requires $\Omega_x = \Omega\cos\Phi$, $\Omega_y = \Omega\sin\Phi$, and $\Omega = |\varepsilon|$. Motzoi et al. write the quadratures as $\mathcal{E}^x$ and $\mathcal{E}^y$. Their $\mathcal{E}$ is this ε, not the noise strength.
- λ is the ratio of the $|1\rangle \leftrightarrow |2\rangle$ coupling to the $|0\rangle \leftrightarrow |1\rangle$ coupling. Motzoi et al. set $\lambda_1 = 1$ and $\lambda_2 = \lambda$. For a transmon, $\lambda = \sqrt{2}$. The SCQC references do not have this parameter because they use a two-level Hilbert space.

The sign of every DRAG quadrature below is tied to Eq. (9). Reversing the sign in $\varepsilon = \Omega_x + \mathrm{i}\Omega_y$, or replacing α by $|\alpha|$ while keeping the same formula, flips those signs. It does not change the geometric closure conditions.

The `qurveros` control dictionary does not store $\varepsilon(t)$ directly. After `evaluate_control_dict`, `times` runs over $s \in [0,1]$, `omega` is a real signed envelope, `phi` is a phase, and `delta` is a detuning. Their relation to the physical fields is

$$\begin{aligned} u(s) &= T\varepsilon(Ts) = \omega_{\mathrm{qv}}(s)\, e^{\mathrm{i}\phi(s)}, \\ d(s) &= T\delta(Ts), \\ \nu &= \alpha T, \\ \eta &= \frac{1}{|\nu|}. \end{aligned} \tag{10}$$

The four symbols on the left are the ones used in the leakage calculation:

- $u(s)$ is the dimensionless complex envelope. Its real and imaginary parts are $\omega_x(s)$ and $\omega_y(s)$.
- $\omega_{\mathrm{qv}}(s)$ is the number stored as `control_dict['omega']`. For a curve without inflection points it equals $T\kappa(t) = T\Omega(t)$. At an inflection it also includes the sign tracker described in Section 2.1. It is not the transition frequency ω_{01}.
- $\phi(s) = \Phi(Ts)$ is `control_dict['phi']`.
- $d(s)$ is `control_dict['delta']`. Because the simulator integrates over $s \in [0,1]$ and the stored `times` array already ends at 1, this number is the physical detuning multiplied by T, not a second copy of that factor.
- ν is the dimensionless anharmonicity. It is the frequency at which the leakage integrals are evaluated. It is not a curvature.

- η is the DRAG expansion parameter. The expansion is appropriate when the envelope and phase vary slowly relative to $|\alpha|$, and it becomes less reliable when $|\alpha|T$ is only a few radians.

Table 1: The same physical quantity in the three sources used here. A dash means that the source does not define the quantity.

This note	`qurveros`	Motzoi et al.	Barnes et al.
$\alpha = \omega_{12} - \omega_{01}$	not used	$\Delta = \omega_2 - 2\omega_1$	$\Delta(t)$ is detuning, not anharmonicity
$\delta(t) = \Delta_c(t)$	`delta` $= d(s)$	δ_1, plus a Stark correction	$\Delta(t)$
$\varepsilon = \Omega_x + \mathrm{i}\Omega_y$	`omega`, `phi`	$\mathcal{E}^x, \mathcal{E}^y$	$\Omega(t), \Phi(t)$
$\kappa = \Omega$	`curvature`; `omega` includes T and the sign tracker	—	$\kappa = \Omega$
$\tau_c = \dot{\Phi} - \Delta_c$	`torsion`	—	torsion τ
θ_B	`barq_angle` in the gate-fixing parameters	—	not part of generic SCQC
δ_z	additive shift of `delta`	—	implicit coefficient of σ_z
$Q = \|2\rangle\langle 2\|$	—	leakage kept in the three-level matrix	Q instead denotes $\delta H/\|\delta H\|$

Remark 1 (Three different statements about leakage)**.** The derivations below keep three statements separate. They use only symbols already defined in this section.

1. *Bare spectral notch.* For an envelope $\varepsilon(t)$ that vanishes at both endpoints, the corrected envelope $\varepsilon - \mathrm{i}\dot{\varepsilon}/\alpha$ has zero Fourier component at angular frequency $-\alpha$. This statement does not involve U_c or the populations in the qutrit.

2. *First-order gate leakage.* The amplitude to arrive in $|2\rangle$ is the integral of $u^*(s)e^{\mathrm{i}\nu s}$ against the ideal qubit amplitude $\langle 1|\tilde{U}_q(s)j\rangle$. Setting those integrals to zero is stronger than a notch in u alone.

3. *Fixed-axis DRAG.* Motzoi et al. remove the first leakage coupling by a frame change $V(t)$ generated by the drive. Their derivation assumes a real envelope $\mathcal{E}^x(t)$ and a corrective quadrature $\mathcal{E}^y(t)$. The complex formula used for a general SCQC phase is the extension of that result to $\varepsilon(t)$, and it remains an ansatz until a qutrit calculation or a simulation checks it. A Schrieffer–Wolff calculation means precisely this elimination of the off-resonant coupling by a near-identity unitary; no additional symbol is introduced by that name.

4 First-order leakage

Let $U_q(t)$ be the ideal evolution generated by the computational part of (9), embedded in the qutrit space, and let $\tilde{U}_q(s) = U_q(Ts)$. Treat the $1 \leftrightarrow 2$ coupling as a perturbation. For a computational input $|j\rangle$, the first-order amplitude to occupy $|2\rangle$ at time T is

$$A_{2j}^{(1)}(T) = -\mathrm{i}\frac{\lambda}{2}\int_0^T \mathrm{d}t\; \varepsilon^*(t)\, e^{\mathrm{i}\alpha t}\, \langle 1|U_q(t)j\rangle. \tag{11}$$

Changing to normalized variables cancels one factor of T:

$$A_{2j}^{(1)} = -\mathrm{i}\frac{\lambda}{2}\int_0^1 \mathrm{d}s\; u^*(s)\, e^{\mathrm{i}\nu s}\, \langle 1|\tilde{U}_q(s)j\rangle. \tag{12}$$

Averaging the resulting probability over the two computational inputs gives

$$\bar{L}^{(1)} = \frac{1}{2} \sum_{j=0,1} |A_{2j}^{(1)}|^2 = \frac{\lambda^2}{8} \sum_{j=0,1} |\ell_j(\nu)|^2, \tag{13}$$

where the leakage moments are

$$\ell_j(\nu) \equiv \int_0^1 \mathrm{d}s \; u^*(s)\, e^{\mathrm{i}\nu s}\, \langle 1|\tilde{U}_q(s)j\rangle. \tag{14}$$

Proposition 1 (Nominal first-order leakage closure)**.** The first-order leakage amplitude vanishes for every computational input if and only if

$$\ell_0(\nu) = 0, \qquad \ell_1(\nu) = 0. \tag{15}$$

These are two complex equations, hence four independent real constraints.

The cheaper proxy $\int_0^1 u(s)e^{\mathrm{i}\nu s}\,\mathrm{d}s = 0$ forces a notch in the bare envelope spectrum. It is not equivalent to (15), because the integrands contain the evolving factor $\langle 1|\tilde{U}_q(s)j\rangle$. That factor is available from the ideal qubit propagator already computed by SCQC, so the stronger condition does not require a qutrit simulation.

If a physical parameter q is uncertain, first-order insensitivity of the leakage cancellation adds

$$\partial_q \ell_0(\nu) = 0, \qquad \partial_q \ell_1(\nu) = 0, \tag{16}$$

another four real constraints for each such parameter. The important choices are anharmonicity, a common amplitude scale, and a static detuning.

5 DRAG as a post-processor

5.1 Fixed-axis transformation

Suppose first that the intended envelope is real. Denote it by $E_x(t)$, and write $E_y(t)$ for a second, corrective quadrature. These are Motzoi's $\mathcal{E}^x$ and $\mathcal{E}^y$; in the notation of Eq. (9), $E_x = \Omega_x$ and $E_y = \Omega_y$ before any correction. The Pauli-like operators that generate the frame change are

$$Y_{jk} = \mathrm{i}|k\rangle\langle j| - \mathrm{i}|j\rangle\langle k|, \tag{17}$$

so Y_{01} acts on the qubit transition and Y_{12} on the leakage transition. This is Motzoi's $\sigma_{j,k}^y$. Following that paper [1], remove the direct leakage coupling by

$$V(t) = \exp\left\{-\mathrm{i}\frac{E_x(t)}{2\alpha}(Y_{01} + \lambda Y_{12})\right\}. \tag{18}$$

The adiabaticity parameter is E_x/α. To first order in that parameter, the transformed Hamiltonian contains the inertial term

$$\frac{1}{2}\left(E_y + \frac{\dot{E}_x}{\alpha}\right)(Y_{01} + \lambda Y_{12}) \tag{19}$$

and an ac-Stark shift of $|1\rangle$ proportional to

$$\delta - \frac{(\lambda^2 - 4)E_x^2}{4\alpha}. \tag{20}$$

Thus the first-order DRAG controls are

$$E_y(t) = -\frac{\dot{E}_x(t)}{\alpha}, \qquad \delta(t) = \frac{(\lambda^2 - 4)E_x(t)^2}{4\alpha}. \tag{21}$$

For a transmon, $\lambda^2 = 2$, so the detuning simplifies to $\delta \approx -E_x^2/(2\alpha)$. Higher-order terms in Ref. [1] correct the in-phase envelope, the derivative quadrature, and the detuning at orders E_x^3/α^2, $E_x^2\dot{E}_x/\alpha^3$, and E_x^4/α^3. They are useful once η is no longer small, but they are not required for the first-order theory.

The frame transformation has no net effect on the gate precisely when

$$V(0) = V(T) = I. \tag{22}$$

For (18), this is the endpoint condition $E_x(0) = E_x(T) = 0$.

5.2 Complex envelope

A general SCQC pulse already uses both quadratures. The covariant extension of (21) is

$$\varepsilon_D(t) = \varepsilon(t) - \mathrm{i}\frac{\beta}{\alpha}\dot{\varepsilon}(t), \tag{23}$$

with a dimensionless calibration parameter β equal to one in the analytic first-order theory. In normalized variables,

$$u_D(s) = u(s) - \mathrm{i}\frac{\beta}{\nu}u'(s). \tag{24}$$

Equivalently, in Cartesian components,

$$\omega_{x,D} = \omega_x + \frac{\beta}{\nu}\omega_y', \qquad \omega_{y,D} = \omega_y - \frac{\beta}{\nu}\omega_x'. \tag{25}$$

If $u = \omega e^{\mathrm{i}\phi}$, the same correction is

$$u_D(s) = e^{\mathrm{i}\phi(s)}\left[\omega(s) + \frac{\beta}{\nu}\omega(s)\phi'(s) - \mathrm{i}\frac{\beta}{\nu}\omega'(s)\right]. \tag{26}$$

The term proportional to $\omega\phi'$ is a radial correction generated by phase modulation. It is absent from textbook fixed-axis DRAG and should not be dropped for an SCQC pulse. Derivatives should be taken on the Cartesian quadratures: near a signed-envelope zero, ω and ϕ separately can have spurious singularities even though ω_x and ω_y are smooth.

Proposition 2 (Bare spectral notch)**.** Suppose $\varepsilon(0) = \varepsilon(T) = 0$. Then

$$\int_0^T \left(\varepsilon(t) - \mathrm{i}\frac{\dot{\varepsilon}(t)}{\alpha}\right) e^{\mathrm{i}\alpha t}\,\mathrm{d}t = 0. \tag{27}$$

Proof. Integrate the derivative term by parts. The boundary term is $(\varepsilon(T)e^{\mathrm{i}\alpha T} - \varepsilon(0))/\alpha = 0$, while $-\mathrm{i}\alpha\int \varepsilon e^{\mathrm{i}\alpha t}\,\mathrm{d}t$ cancels the remaining integral of $\varepsilon e^{\mathrm{i}\alpha t}$. □

Proposition 2 does not imply Proposition 1. The latter weights $\varepsilon^*(t)$ by $\langle 1|U_q(t)j\rangle$, and DRAG also changes U_q at order η. Equation (23) should therefore be read as the appropriate post-processing ansatz, followed by a small calibration of β, an overall amplitude, and a constant detuning.

5.3 Boundary conditions and calibration

The following conditions are listed in the order they become necessary.

1. $u(0) = u(1) = 0$ makes the first-order DRAG frame an identity at both endpoints. BARQ already encourages this through its vanishing-envelope construction, but a loaded or rescaled pulse should be checked.

2. $u'(0) = u'(1) = 0$ makes the added derivative quadrature start and end continuously.

3. If corrections through the mth derivative are used, the derivatives of orders $0, \dots, m$ should vanish at both endpoints. Otherwise integration by parts leaves boundary kicks in the leakage amplitude.

4. After constructing u_D, at least the global amplitude and a constant phase or detuning offset should be recalibrated. The derivative term does not cancel every Stark and higher-order coherent error.

A practical post-processing sequence is:

1. Design $u(s)$ and $d(s)$ with ordinary BARQ and record the original closure and tantrix-area diagnostics.

2. Apply (24) and the Stark correction from (21), transformed into normalized units.

3. On a qutrit model, tune β, the amplitude scale, and a constant detuning to minimize $1-F_{\text{avg}}$ and the leakage L defined in Section 7.2.

4. Recompute noise generators with the corrected propagator, rather than reusing the original curve tests.

5. Repeat under variation of α, gain, timing, sampling, and the control-line transfer function.

6 Robustness after DRAG

6.1 Generator conditions

The original geometric tests are statements about a qubit propagator and a qubit noise operator. Both change once the pulse is corrected and embedded in a qutrit. Let $U_D(t)$ be the ideal qutrit evolution generated by the corrected controls. For a parameter q_a with Hamiltonian derivative $V_a(t) = \partial H/\partial q_a$, the first Magnus generator is

$$G_a = \int_0^T \mathrm{d}t\, U_D^\dagger(t)\, V_a(t)\, U_D(t). \tag{28}$$

Its computational and leakage blocks impose different requirements.

Definition 1 (First-order qutrit robustness)**.** The corrected gate is first-order robust against q_a when

$$PG_aP - \frac{1}{2}\,\mathrm{Tr}(PG_aP)\,P = 0, \qquad QG_aP = 0. \tag{29}$$

The first equation removes the traceless logical rotation. The second prevents the noise from creating leakage to first order.

For a genuine qubit model one has $P = I$, and the first equation reduces to the usual closure condition. Applying the old closure or tantrix test to the uncorrected curve is not sufficient, because neither U_D nor V_a is the operator assumed by that test.

6.2 Perturbation of dephasing closure

Let U_0 be the computational evolution generated by the uncorrected SCQC pulse, embedded in the qutrit space. The corrected propagator can be factored as

$$U_D = U_0 \exp[-\mathrm{i}\eta K_1 + \mathcal{O}(\eta^2)], \tag{30}$$

where the Hermitian operator $K_1(t)$ collects every first-order change produced by the derivative quadrature and by the frame $V(t)$. Its explicit matrix is not required for the scaling below;

only the existence of a bounded K_1 at order η is used. For quasistatic detuning noise, take the interaction-picture operator $n_0(t) = U_0^\dagger(t)\sigma_z U_0(t)$ and the associated error curve $r_0(T) = \int_0^T n_0(t)\,\mathrm{d}t$. Expanding (28) gives

$$r_D(T) = r_0(T) + \mathrm{i}\eta \int_0^T [K_1(t), n_0(t)]\,\mathrm{d}t + \mathcal{O}(\eta^2). \tag{31}$$

Ordinary SCQC can enforce $r_0(T) = 0$. It does not enforce the commutator integral. Consequently, a pulse that is first-order dephasing-robust before post-processing generically acquires a residual generator of order η.

For a qubit, a vanishing first-order generator changes the leading coherent infidelity from quadratic to quartic in the noise strength. After unconstrained post-processing the scaling is instead

$$\begin{aligned} \text{original closure:} \quad & 1 - F = \mathcal{O}\big((\delta_z T)^4\big), \\ \text{post-processed DRAG:} \quad & 1 - F = \mathcal{O}\big((\delta_z T)^2 \eta^2\big) + \mathcal{O}\big((\delta_z T)^4\big). \end{aligned} \tag{32}$$

The mixed term dominates the original quartic protection whenever $|\delta_z T| \lesssim \eta$. This is a scaling statement, not a claim that the numerical coefficient is large. Restoring the quartic regime requires the additional vector condition

$$\int_0^T [K_1(t), n_0(t)]\,\mathrm{d}t = 0, \tag{33}$$

or, more directly, the block conditions (29) evaluated on the final pulse.

6.3 Amplitude, DRAG gain, and anharmonicity

If one calibration error a multiplies both applied quadratures, the physical corrected quadratures are $\Omega_{x,D}(t) = \omega_{x,D}(t/T)/T$ and $\Omega_{y,D}(t) = \omega_{y,D}(t/T)/T$. The Hamiltonian derivative with respect to that gain is

$$V_a(t) = \frac{1}{2}\big(\Omega_{x,D}(t)\,\sigma_x + \Omega_{y,D}(t)\,\sigma_y\big), \tag{34}$$

and robustness is $G_a \propto I$ on the computational block. The original tantrix-area condition is the special case of this statement before DRAG and before the qutrit embedding. If the derivative channel is calibrated independently, β must be treated as its own parameter and G_β must vanish separately. Anharmonicity uncertainty is qualitatively different: it moves the leakage phase $e^{\mathrm{i}\nu s}$ and the coefficient $1/\nu$ in (24) simultaneously. A first-order robust design therefore penalizes $\partial_\alpha M$, where $M = PU_DP$ is the final computational block.

Second-order cancellation follows from the next Magnus term. For two error channels,

$$G_{ab}^{(2)} = -\frac{\mathrm{i}}{2}\int_0^T \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\ [V_a^I(t_1), V_b^I(t_2)], \tag{35}$$

where the superscript I denotes the interaction picture generated by U_D. The computational block of $G_{ab}^{(2)}$, with its trace removed, is the appropriate second-order condition. In the original qubit dephasing model this is the geometric zero-area constraint.

7 Leakage-aware BARQ

Post-processing takes a curve designed for a qubit and changes its pulse. Leakage-aware BARQ instead varies the free Bézier points, and optionally a few gate-fixing parameters, under a qutrit objective. Three nested models are useful.

Table 2: Noise channels that should be recomputed after DRAG.

Uncertainty	Generator	New risk
Static detuning	$\partial H/\partial\delta$	Closure shifted at $\mathcal{O}(\eta)$
Common I/Q gain	$\partial H/\partial a$	Derivative terms scale with the gain
DRAG coefficient	$\partial H/\partial\beta$	Separate derivative-channel error
Anharmonicity	$\partial H/\partial\alpha$	Notch frequency and $1/\alpha$ both move
Transfer function	filtered functional derivative	Differentiation amplifies distortion

7.1 Spectral and perturbative objectives

The least expensive differentiable objective uses Proposition 1:

$$J_{\mathrm{spec}} = \sum_{j=0,1} |\ell_j(\nu)|^2. \tag{36}$$

The bare-envelope proxy $J_{\mathrm{bare}} = |\int_0^1 u(s)e^{\mathrm{i}\nu s}\,\mathrm{d}s|^2$ is cheaper but weaker. The corresponding leakage probability from (13) is

$$J_{\mathrm{leak},1} = \frac{\lambda^2}{8}(|\ell_0|^2 + |\ell_1|^2). \tag{37}$$

Both can be evaluated from the normalized envelope and the ideal qubit propagator. Their gradients exist once the cumulative phase and arclength are computed by a differentiable quadrature rather than by a black-box integrator.

7.2 Exact qutrit objective

Let U be the full qutrit propagator and $M = PUP$ its 2×2 computational block. For a pure input drawn uniformly from the qubit Bloch sphere, the average probability of leaving that subspace is

$$L = 1 - \frac{1}{2}\operatorname{Tr}(M^\dagger M). \tag{38}$$

If U_g is the target qubit unitary, the Haar-averaged state fidelity is

$$F_{\mathrm{avg}} = \frac{\operatorname{Tr}(M^\dagger M) + |\operatorname{Tr}(U_g^\dagger M)|^2}{6}. \tag{39}$$

Equation (39) follows from

$$\int |\langle\psi|A|\psi\rangle|^2\,\mathrm{d}\psi = \frac{|\operatorname{Tr} A|^2 + \operatorname{Tr}(A^\dagger A)}{d(d+1)} \tag{40}$$

with $d = 2$ and $A = U_g^\dagger M$. The leakage term (38) is already present in F_{avg}, but retaining it separately allows leakage to be weighted more heavily than a coherent logical rotation. Both quantities remain meaningful when M is not unitary.

7.3 What remains analytic

BARQ's point-gate-fixing construction fixes the ideal qubit operation and the endpoint contribution to dephasing closure before numerical optimization. On a qutrit, Stark shifts and virtual transitions modify M, so those constraints no longer guarantee the physical target exactly. The useful hybrid is:

- retain point-gate fixing, so the zeroth-order qubit gate and curve closure remain exact;

- constrain enough endpoint control points that u and u' vanish at $s = 0, 1$;
- optimize interior points against F_{avg}, L, and the corrected generators (29);
- release a global amplitude and a constant detuning during the final stage, since qutrit gate fixing is no longer purely analytic;
- test four- and five-level truncations for short gates, where two-photon and higher-state transitions are outside (9).

A concrete staged objective is

$$\begin{aligned} J = {} & w_g(1 - F_{\text{avg}}) + w_L L + w_z \|PG_zP - \tfrac{1}{2}\operatorname{Tr}(PG_zP)P\|_F^2 + w_{zL}\|QG_zP\|_F^2 \\ & + w_a \|PG_aP - \tfrac{1}{2}\operatorname{Tr}(PG_aP)P\|_F^2 + w_{aL}\|QG_aP\|_F^2 + w_\alpha \|\partial_\alpha M\|_F^2 \\ & + w_b \int_0^1 |u'(s)|^2 \,\mathrm{d}s + w_{bb} \int_0^1 |u''(s)|^2 \,\mathrm{d}s + w_{\max} \max_s |u(s)| \\ & + w_{\text{end}} \sum_{k=0,1} \sum_{s_\star \in \{0,1\}} |u^{(k)}(s_\star)|^2. \end{aligned} \tag{41}$$

Every w is a nonnegative weight chosen by the user, not a physical constant. G_z and G_a are the generators (28) for δ_z and for the common gain a. The weights should be introduced in stages: target and original robustness, then nominal leakage, then parameter uncertainty, and finally bandwidth and amplitude. Turning every term on simultaneously makes the trade-off difficult to diagnose.

8 Further extensions

8.1 Choosing the gate duration

For a fixed normalized BARQ waveform, the first-order leakage depends on duration only through $\nu = \alpha T$. A coherent-error search can therefore minimize $L_1(\nu) = J_{\text{leak},1}$ over T, subject to the hardware bound on peak amplitude. This is stronger than placing a zero of the bare spectrum at α, because ℓ_j contains $\tilde{U}_q$. Decoherence selects a finite optimum of a model such as

$$E(T) \approx L_{\text{leak}}(\alpha T) + \frac{c_1 T}{T_1} + \frac{c_\phi T}{T_\phi} + E_{\text{control}}(T). \tag{42}$$

Here T_1 and T_ϕ are the relaxation and pure-dephasing times, c_1 and c_ϕ are dimensionless averages that depend on the target gate, L_{leak} is the coherent leakage probability, and E_{control} collects coherent control error that remains after leakage is removed. The coefficients should come from an open-system simulation rather than from a universal rule.

8.2 Endpoint smoothness

Let $I(\nu) = \int_0^1 g(s) e^{\mathrm{i}\nu s} \,\mathrm{d}s$. If g and its first $m-1$ derivatives vanish at both endpoints, repeated integration by parts yields

$$I(\nu) = \left(-\frac{1}{\mathrm{i}\nu}\right)^m \int_0^1 g^{(m)}(s) e^{\mathrm{i}\nu s} \,\mathrm{d}s, \tag{43}$$

and therefore $|I| = \mathcal{O}(|\nu|^{-m})$. For leakage, $g(s) = u^*(s)\langle 1|\tilde{U}_q(s)j\rangle$. Making only u smooth does not automatically make every derivative of this product vanish, but it improves the decay. The quadratic penalties on u' and u'' in (41) suppress bandwidth even when an endpoint derivative cannot be constrained exactly.

8.3 Higher-derivative and multi-transition DRAG

A linear differential filter can place spectral zeros at unwanted angular frequencies Δ_k. Each Δ_k is a transition frequency in radians per unit time, such as α or a spectator splitting. It is not the control detuning Δ_c. Starting from a base dimensionless envelope $u_0(t)$,

$$u_{\mathrm{corr}}(t) = \prod_k \left(1 - \frac{\mathrm{i}}{\Delta_k}\frac{\mathrm{d}}{\mathrm{d}t}\right) u_0(t). \tag{44}$$

Under the Fourier convention $\int u(t)e^{\mathrm{i}\omega_F t}\,\mathrm{d}t$, where ω_F is the Fourier angular frequency and not a Rabi frequency, the kth factor contributes $1 - \omega_F/\Delta_k$. If the filter is instead applied to the normalized envelope $u(s)$, replace $\mathrm{d}/\mathrm{d}t$ by $T^{-1}\mathrm{d}/\mathrm{d}s$ and Δ_k by the dimensionless frequency $\Delta_k T$. Expanding (44) produces the higher-derivative DRAG family. The same construction suppresses several internal or spectator transitions, provided the corresponding endpoint derivatives vanish and the coefficients are recalibrated on the full propagator. At very short duration, direct and multiphoton transitions beyond $|2\rangle$ can dominate the residual error, so the filter frequencies must include those transitions or the simulation must retain more levels.

8.4 Fourier notches and Wah-Wah modulation

The derivative expansion is not the only way to impose the linear constraints $\ell_j(\nu_k) = 0$. Choose real basis functions $f_n(s)$ on $[0, 1]$, for example a Fourier subset, and write $u(s) = \sum_n c_n f_n(s)$ with complex coefficients c_n. Each notch becomes a linear condition on those coefficients. The conditions can be combined with the BARQ gate area, endpoint, and robustness constraints. Wah-Wah control is the crowded-spectrum version of this idea: sideband modulation places a notch near a neighbour's leakage transition, while a DRAG quadrature suppresses the addressed qubit's own $1 \leftrightarrow 2$ transition [3, 4]. It is relevant when a spectator transition is closer to the drive than the addressed qubit's anharmonicity.

8.5 Relation to multilevel SCQC

The generalized geometric formalism represents an interaction-picture operator by coordinates on a generator basis. Let Λ_a, with $a = 1, \ldots, 8$ for a qutrit, be Hermitian generators normalized by $\mathrm{Tr}(\Lambda_a\Lambda_b) = 2\delta_{ab}$. These are the Gell-Mann matrices in the qutrit case and the Pauli matrices when the same formula is reduced to a qubit. For an operator Q_{B} as in Eq. (5),

$$q_a(t) = \frac{1}{2}\,\mathrm{Tr}\big[\Lambda_a U_c^\dagger(t) Q_{\mathrm{B}} U_c(t)\big]. \tag{45}$$

The symbol q_a is a coordinate, not the uncertain parameter q used earlier to denote a generic derivative. Leading-order cancellation means that every coordinate closes, $\int_0^T q_a(t)\,\mathrm{d}t = 0$. A generic qutrit error has eight coordinates, so the curve is eight-dimensional. Generalized curvatures can be connected recursively to the controls, but a transmon does not supply enough independent controls to realize an arbitrary curve: two quadratures and one detuning act with linked $0 \leftrightarrow 1$ and $1 \leftrightarrow 2$ matrix elements.

The perturbative intermediate theory avoids that inverse problem. Let $\mathcal{H}$ be the three-dimensional state space. Leakage amplitudes live in the four-real-dimensional space $Q\mathcal{H}P \cong \mathbb{C}^2$, and the two components of its integrated coordinate are exactly (ℓ_0, ℓ_1). Closure of this leakage curve is Proposition 1. Thus leakage-aware BARQ adds a second, four-dimensional closure condition without abandoning the existing SU(2) curve used to design the computational gate.

9 Expected improvement and required evidence

For a fixed rotation angle, the drive amplitude scales as $1/T$. The dimensionless expansion parameter is therefore $\eta \sim 1/(|\alpha|T)$, up to a pulse-shape coefficient. A generic off-resonant

leakage amplitude begins at $\mathcal{O}(\eta)$, so its probability begins at $\mathcal{O}(\eta^2)$. First-order DRAG is constructed to cancel that amplitude. The residual coherent leakage is then commonly $\mathcal{O}(\eta^2)$ in amplitude and $\mathcal{O}(\eta^4)$ in probability, before decoherence and model error are added. Table 3 gives these powers with unit prefactors. It is a regime guide, not a predicted infidelity.

Table 3: Unit-prefactor scaling for $|\alpha|/2\pi = 200\,\mathrm{MHz}$. The entries are η^2 and η^4, not calculated gate errors.

T (ns)	$\lvert\nu\rvert = \lvert\alpha\rvert T$	η	$\mathcal{O}(\eta^2)$	$\mathcal{O}(\eta^4)$
6.25	7.85	0.127	1.62×10^{-2}	2.63×10^{-4}
10	12.57	0.0796	6.33×10^{-3}	4.01×10^{-5}
20	25.13	0.0398	1.58×10^{-3}	2.51×10^{-6}
40	50.27	0.0199	3.96×10^{-4}	1.57×10^{-7}

Published results show that the attainable improvement depends strongly on duration and pulse family. Motzoi et al. obtain reductions from roughly one to several orders of magnitude in simulation, depending on the waveform [1]. Early transmon experiments demonstrated derivative-of-Gaussian DRAG at few-nanosecond duration [2]. At 4.16 ns, closed-loop optimization around DRAG reduced leakage from 0.29% to 0.044% and raised the average Clifford fidelity from 99.11% to 99.76% on that device [5]. Analytical FAST and higher-derivative DRAG have subsequently reached leakage below 3×10^{-5} at 6.25 ns, about a twentyfold reduction relative to a compared cosine-DRAG pulse [8]. None of these values can be transferred directly to a BARQ waveform: its spectrum, phase excursions, anharmonicity, and coherence environment are different. Experimental SCQC benchmarking also identifies leakage as a gate-time-dependent limitation and selects duration by balancing it against decoherence [9].

A numerical claim for this package needs the comparisons in Table 4. The minimum publication-quality comparison is a qutrit simulation of the original BARQ pulse, the post-processed pulse, and a pulse reoptimized with (41), all evaluated at the same peak amplitude and coherence times.

Table 4: Simulation campaign needed before quoting an improvement.

Sweep	Quantities	Purpose
T and α	F_{avg}, L, logical error	Locate leakage minima and the perturbative regime
β and Stark coefficient	two-dimensional error surface	Separate leakage cancellation from phase correction
Static δ_z and gain	infidelity heat maps	Test whether robustness survives post-processing
Three to five levels	convergence of F_{avg} and L	Detect higher-level and multiphoton leakage
T_1, T_ϕ	open-system process fidelity	Choose a useful duration
Sampling and transfer function	filtered-pulse fidelity	Test sensitivity of derivative controls
α and gain mismatch	worst-case and average fidelity	Quantify calibration robustness

10 Conclusion

DRAG fits SCQC as a correction in the pulse domain. For a complex BARQ envelope the first-order post-processor is (24), supplemented by a quadratic Stark detuning and by the endpoint conditions $u = u' = 0$. This exactly notches the bare envelope spectrum, but it only approximately preserves the computational gate and the original closure condition. The missing

robustness requirement is (29), evaluated with the corrected qutrit evolution; its first correction beyond ordinary curve closure is the mixed integral (33).

Leakage-aware BARQ imposes the two complex constraints $\ell_0(\alpha T) = \ell_1(\alpha T) = 0$ and can include their derivatives with respect to uncertain parameters. Exact qutrit propagation remains the acceptance test, with L and F_{avg} given by (38) and (39). Gate-time selection, endpoint smoothness, higher-derivative filters, and spectral notches are compatible extensions. A full eight-dimensional SU(3) construction is consistent with the general geometric formalism, whereas the practical and geometrically faithful intermediate step is closure of the perturbative leakage curve in $\mathbb{C}^2$.

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